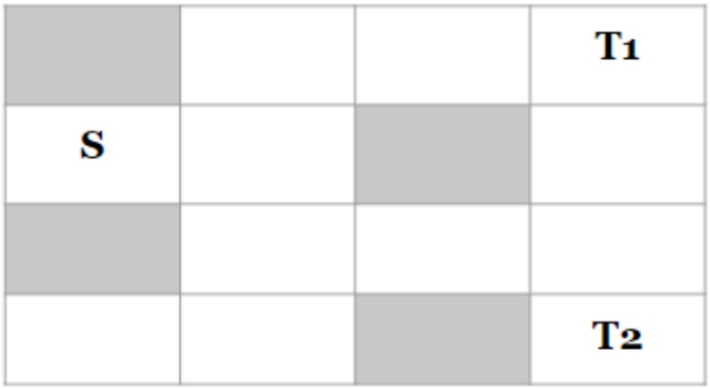
Previous Years Question

Routing Algorithm

Q-1:

For the 4 × 4 maze below, the dark regions are obstacles or components.

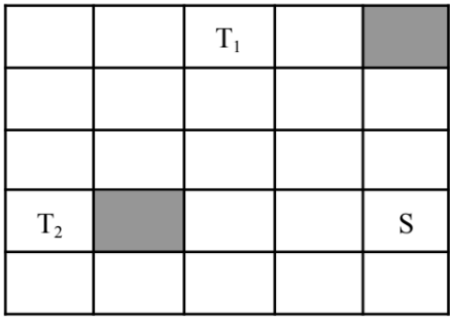


The total memory requirements for the maze is **4 bytes**.

(a)	**Calculate** the number of bits per cell (n) and the maximum value that a cell can store (L) as per the memory requirements	**2+1**
(b)	**State** the sequence for wave propagation as per the memory requirements. Perform wave propagation to find the shortest path from source, **S** to both of the targets, $\mathbf{T_1}$ and $\mathbf{T_2}$	**2+6**
(c)	**Show** the shortest path for each of the targets.	**4**

Q-2:

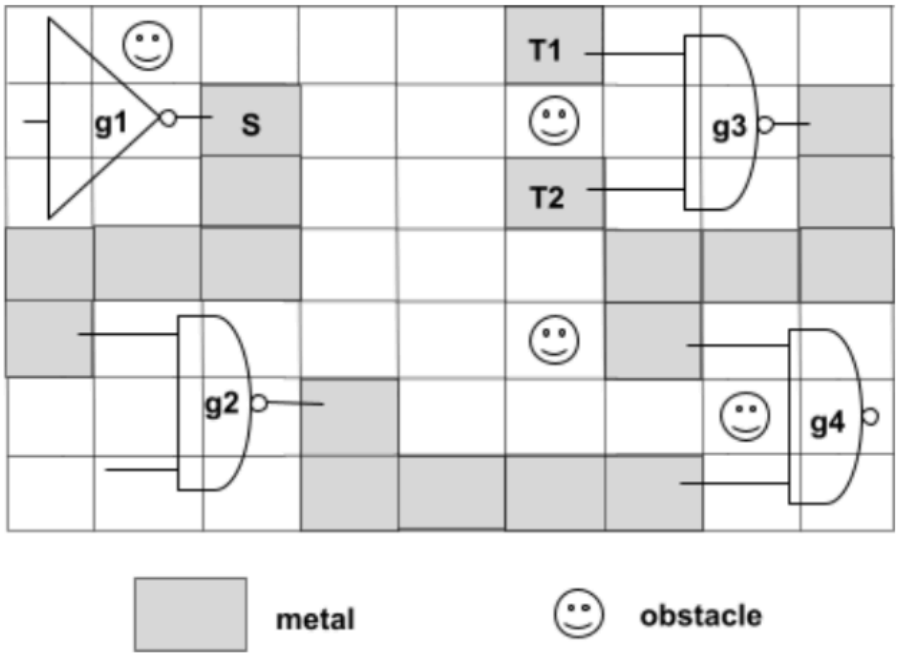
Lee's Maze algorithm can be used to find the shortest path between two nodes avoiding obstacles. Dark regions are obstacles or components.



Find the shortest path from source, S to both the targets, T_1 and T_2. Rule of thumb should be taken into consideration.	**5**

Q-3:

In the following figure, you can see the grids for Lee's Maze Algorithm.



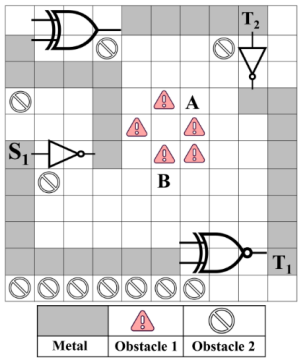
(a)	Using the algorithm, **find** the <u>shortest</u> path from S to T1 and T2 allowing minimum bends. **Answer this question on the next page of the question paper.**	**6**
(b)	**Show** the graph representation of the connected gates ignoring the grids. (Convert gates to nodes).	**3**
(c)	**Compare** the initial cut cost with that of after the first iteration of the KL algorithm for partitioning. Assume initial Cut Sets: A (g1,g2) and B (g3,g4).	**6**

Q-4:

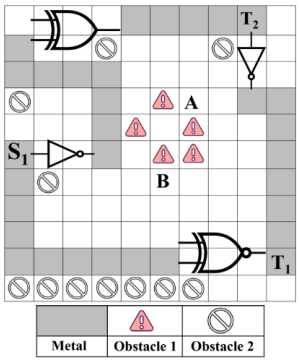
For question (a) you are given a certain chip planning task. You have four blocks with the following widths and heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 3, h = 3 or w = 1, h = 9
For questions (b-e) refer to the figures at the bottom of the question. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown.

(a)	**Find** and draw a floor plan with the minimum total area enclosed by the global bounding box using 10x10 space.	**[3]**
(b)	**Determine** the minimum Manhattan distance from A to B (You don't have to show in grid. Just write the answer).	**[1]**
(c)	Using the Lee's Maze Algorithm, **find** the shortest path from **S1** to **T1** and **T2** *allowing minimum bends*. **Answer this question on this and the subsequent grids on the bottom of the question paper. A and B are not obstacles.**	**[4]**
(d)	**Find** the total memory usage of the given grid if the sequence is a series of natural numbers. If the maximum memory requirement of the provided grid is 27 bytes, calculate the maximum number of bits that each cell can hold.	**[2+3]**
(e)	**Find** the number of consecutive and unique numbers (starting from 0) that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint in (d). What should be a full sequence for wave propagation? **[2]**	

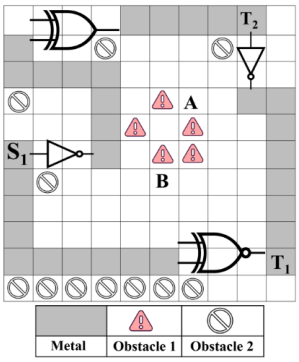
Grid 1



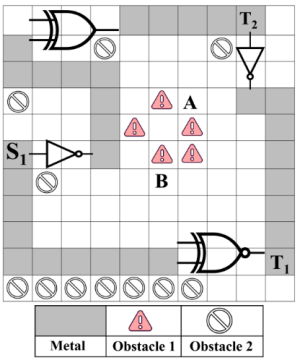
Grid 2



Grid 3



Grid 4

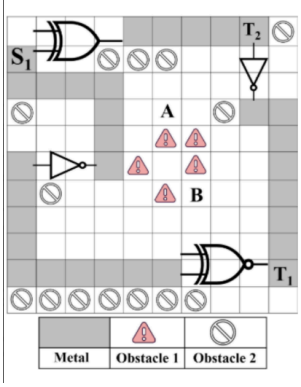


Q-5:

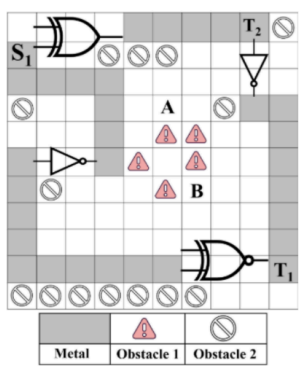
For question (a) you are given a certain chip planning task. You have three blocks with the following widths and heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 3, h = 3 or w = 1, h = 9
For questions (b-e) refer to the adjacent figures at the bottom of the questionr. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown.

(a)	**Find** and draw a floor plan with the minimum total area enclosed by the global bounding box using 10x10 space.	**[3]**
(b)	**Determine** the minimum Manhattan distance from A to B (You don't have to show in grid. Just write the answer).	**[1]**
(c)	Using the Lee's Maze Algorithm, **find** the shortest path from **S1** to **T1** and **T2** *allowing minimum bends*. **Answer this question on this and the subsequent grids on the question paper. A and B are not obstacles.**	**[4]**
(d)	**Find** the total memory usage of the given grid if the sequence is a series of natural numbers. If the maximum memory requirement of the provided grid is 27 bytes, calculate the maximum number of bits that each cell can hold.	**[2+3]**
(e)	**Find** the number of consecutive and unique numbers (starting from 0) that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint in (d). What should be a full sequence for wave propagation? **[2]**	

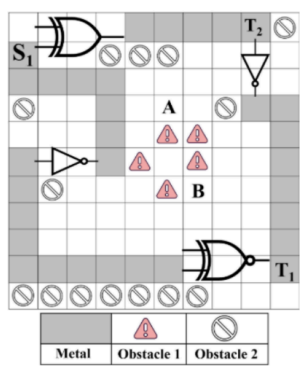
Grid 1



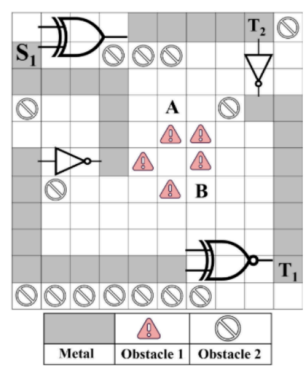
Grid 2



Grid 3

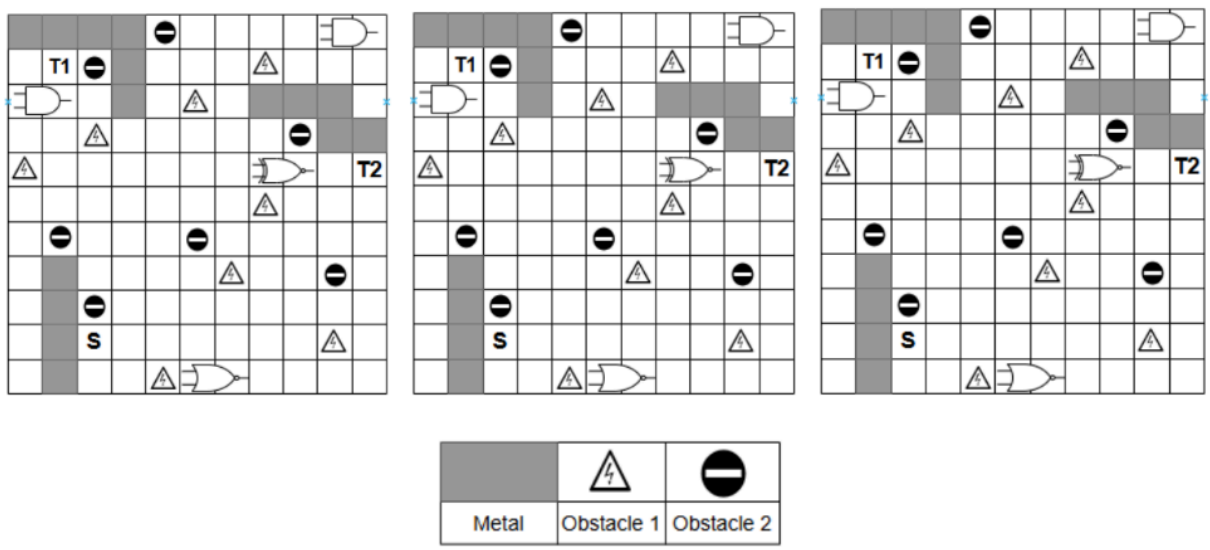


Grid 4



Q-6:

For question (a), you are given a certain chip planning task. You have four blocks with the following widths and
heights:
Block A: w = 1, h = 5 or w = 5, h = 1
Block B: w = 1, h = 3 or w = 3, h = 1
Block C: w = 2, h = 1 or w = 1, h = 2
Block D: w = 9, h = 1 or w = 3, h = 3
For questions (b-d), refer to the figures at the bottom of the question. You can see the grid layout of a system with a source (S1) and two targets – T1 and T2. The obstacles and free cells are shown. **Gates are not obstacles**.



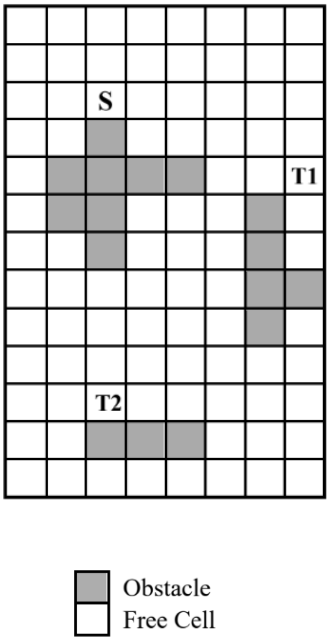
(a)	**Find** and **draw** a floor plan with the minimum total area enclosed if the global bounding box is a 10x10 grid.	**[3]**
(b)	Using the modified Lee's Maze Algorithm, **find** the shortest path from S to T1 and T2, allowing minimum bends. (Use the sequence 123123…….)	**[4]**
(c)	**Find** the total memory usage of the given grid for the sequence mentioned in (b).	**[4]**
(d)	If the maximum memory requirement of the provided grid is 32 bytes, **find** how many unique numbers starting from 0 can be used for wave propagation in Lee's maze algorithm and suggest a full sequence.	**[4]**

Q-7:

For question (a) you are given a certain **chip planning task.** You have three blocks with the following widths and heights:

Block A: $w = 2, h = 3$ or $w = 3, h = 2$
Block B: $w = 1, h = 2$ or $w = 2, h = 1$
Block C: $w = 1, h = 4$ or $w = 4, h = 1$

For questions (b-e) refer to the adjacent figures on the right of the paper, you can see the grid layout of a system with a source (**S**) and two targets – **T1** and **T2.** The obstacles and free cells are shown.



(a) **Define** chip planning. **Find** and draw a floorplan with the minimum total area enclosed by its global bounding box. **[3]**

(b) Using the Lee's Maze Algorithm, **find** the shortest path from **S** to **T1** and **T2** *allowing minimum bends*. **[4]**
Answer this question on the grids on the question paper.

(c) **Find** the total memory usage of the given grid if the sequence is a series of natural numbers. **[3]**

(d) If the maximum memory requirement of the provided grid is **36 bytes**, **calculate** the maximum number of bits that each cell can hold. **[3]**

(e) **Find** the number of consecutive and unique numbers (starting from 1) that can be used for wave propagation in the Lee's Maze algorithm, for the given memory constraint. **[2]**

What should be a full sequence for wave propagation?